Brutal Practice Paper B18

Each question has 6 to 8 options; exactly ONE is correct. Answers are scattered - there is NO pattern. Every question needs several steps - read carefully and work it out fully.

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1. A torch uses two cells of 1.5 V each joined in series. A bigger lamp needs at least 6 V to glow brightly. How many such 1.5 V cells, joined in series, will the bigger lamp need? [Electric Current & its Effects]

- **(A)** 4 cells
- **(B)** 6 cells
- **(C)** 2 cells
- **(D)** 12 cells
- **(E)** 5 cells
- **(F)** 8 cells

2. Work out (-3/4) + (5/6) and give the answer in lowest terms. [Rational Numbers]

- **(A)** 1/12
- **(B)** -1/12
- **(C)** -1/2
- **(D)** 2/10
- **(E)** 19/12
- **(F)** 8/12

3. In a lab, 30 solutions are tested. 12 of them turn blue litmus red, and 8 of them turn red litmus blue. The rest change neither litmus paper. How many of the solutions are neutral? [Acids, Bases & Salts]

- **(A)** 8
- **(B)** 4
- **(C)** 18
- **(D)** 22
- **(E)** 10
- **(F)** 12

4. What is the complement of an angle of 37 degrees? [Lines & Angles]

- **(A)** 53 degrees
- **(B)** 143 degrees
- **(C)** 63 degrees
- **(D)** 90 degrees
- **(E)** 43 degrees
- **(F)** 127 degrees
- **(G)** 23 degrees

5. A heating coil warms water at a steady rate. In 4 minutes it raises the temperature by 20 C. If it keeps heating at the same rate, how many degrees will it raise the temperature in 9 minutes? [Electric Current & its Effects]

- **(A)** 40 C
- **(B)** 25 C
- **(C)** 80 C
- **(D)** 20 C
- **(E)** 29 C
- **(F)** 36 C
- **(G)** 45 C

6. Multiply (-2/5) x (15/-8) and simplify. [Rational Numbers]

- **(A)** 3/4
- **(B)** 6/13
- **(C)** -4/3
- **(D)** 17/40
- **(E)** 30/40
- **(F)** 4/3

7. Which of these indicator results is CORRECT? [Acids, Bases & Salts]

- **(A)** China rose gives no colour at all with acid
- **(B)** Turmeric turns red in an acidic solution
- **(C)** Turmeric turns red in a basic solution
- **(D)** Phenolphthalein turns blue in a base
- **(E)** Blue litmus turns blue in an acid
- **(F)** Red litmus turns red in a base
- **(G)** Phenolphthalein turns pink in an acid

8. A pair of complementary angles has one angle equal to 4 times the other angle. Find the larger angle. [Lines & Angles]

- **(A)** 144 degrees
- **(B)** 45 degrees
- **(C)** 80 degrees
- **(D)** 72 degrees
- **(E)** 18 degrees
- **(F)** 60 degrees
- **(G)** 36 degrees

9. Three identical bulbs are joined in a single series loop with a battery. The filament of the middle bulb suddenly breaks. What happens to the other two bulbs, and why? [Electric Current & its Effects]

- **(A)** They go off, because the break leaves the single path with a gap so no current flows
- **(B)** They glow brighter, because more current now flows
- **(C)** Only the bulb nearest the break goes off
- **(D)** They stay the same, because each bulb has its own separate path
- **(E)** They glow dimmer but keep glowing
- **(F)** They glow only after the broken bulb is replaced
- **(G)** They blink on and off again and again

10. Which of these rational numbers lies between 1/3 and 1/2? [Rational Numbers]

- **(A)** 1/5
- **(B)** 5/12
- **(C)** 3/4
- **(D)** 5/6
- **(E)** 7/12
- **(F)** 2/3
- **(G)** 1/4

11. One antacid tablet neutralises 5 mL of stomach acid. A patient has 35 mL of excess acid. How many tablets are needed to neutralise it all? [Acids, Bases & Salts]

- **(A)** 35 tablets
- **(B)** 30 tablets
- **(C)** 6 tablets
- **(D)** 8 tablets
- **(E)** 7 tablets
- **(F)** 5 tablets

12. What is the supplement of an angle of 118 degrees? [Lines & Angles]

- **(A)** 82 degrees
- **(B)** 72 degrees
- **(C)** 152 degrees
- **(D)** 28 degrees
- **(E)** 62 degrees
- **(F)** 118 degrees
- **(G)** 48 degrees

13. A fuse on a wire melts if the current crosses 5 A. Four appliances run together on this wire and each draws 1.2 A. By how many amperes is the wire current below the melting limit of the fuse? [Electric Current & its Effects]

- **(A)** 1.2 A
- **(B)** 4.8 A
- **(C)** 5 A
- **(D)** 9.8 A
- **(E)** 3.8 A
- **(F)** 0.2 A
- **(G)** 0.8 A

14. What is the reciprocal of (-7/9)? [Rational Numbers]

- **(A)** -9/7
- **(B)** 16/63
- **(C)** -7/9
- **(D)** -16/63
- **(E)** 9/7
- **(F)** 0
- **(G)** 7/9

15. When a stinging ant bites, it leaves behind formic acid that burns. Which home item best soothes it, and why? [Acids, Bases & Salts]

- **(A)** Lemon juice, because it is sour
- **(B)** Cold milk, because it is acidic
- **(C)** Vinegar, because it is acidic
- **(D)** Plain salt, because it is neutral
- **(E)** Sugar, because it tastes sweet
- **(F)** More formic acid, to cancel it out
- **(G)** Baking soda, because it is a mild base that neutralises the acid

16. Two angles form a linear pair, and one is 30 degrees more than the other. What are the two angles? [Lines & Angles]

- **(A)** 70 degrees and 110 degrees
- **(B)** 75 degrees and 75 degrees
- **(C)** 75 degrees and 105 degrees
- **(D)** 80 degrees and 100 degrees
- **(E)** 60 degrees and 90 degrees
- **(F)** 65 degrees and 95 degrees
- **(G)** 90 degrees and 120 degrees

17. A scrapyard crane carries an electromagnet that lifts 60 kg of iron scrap each trip and makes 15 trips every hour. How many kilograms of scrap does it move in a 6-hour working shift? [Electric Current & its Effects]

- **(A)** 5040 kg
- **(B)** 5400 kg
- **(C)** 81 kg
- **(D)** 900 kg
- **(E)** 90 kg
- **(F)** 360 kg
- **(G)** 540 kg

18. Find the value of (5/6) - (-2/3). [Rational Numbers]

- **(A)** 7/6
- **(B)** 5/2
- **(C)** 3/2
- **(D)** 1/2
- **(E)** 3/6
- **(F)** -3/2

19. To neutralise acidic waste, a factory mixes base and acid in the ratio 2 : 3. To treat 45 L of acid, how many litres of base are required? [Acids, Bases & Salts]

- **(A)** 30 L
- **(B)** 90 L
- **(C)** 15 L
- **(D)** 67.5 L
- **(E)** 18 L
- **(F)** 45 L

20. When two straight lines cross, the pair of angles directly opposite each other are always what? [Lines & Angles]

- **(A)** Adding to 360 degrees
- **(B)** Always right angles
- **(C)** Supplementary
- **(D)** Always unequal
- **(E)** Complementary
- **(F)** Always adding to 180 degrees
- **(G)** Equal

21. A pupil draws a circuit with one cell, one bulb and a switch that is left OPEN. With the switch open, will the bulb glow, and why? [Electric Current & its Effects]

(A) No, because the wire is too thin

(B) No, because the cell must be empty

(C) No, because the bulb has no filament

(D) No, an open switch leaves a gap so current cannot complete the loop

(E) Yes, but the bulb glows only dimly

(F) Yes, the switch only sets the brightness

(G) Yes, the cell pushes current across the gap

22. Divide (-4/9) by (8/-3) and simplify. [Rational Numbers]

(A) 6

(B) -1/6

(C) 1/6

(D) 32/27

(E) 1/2

(F) 2/3

(G) -6

23. Which of these is a SALT formed by the reaction of an acid with a base? [Acids, Bases & Salts]

(A) Sodium hydroxide

(B) Milk of magnesia

(C) Soap solution

(D) Lemon juice

(E) Sodium chloride (common table salt)

(F) Hydrochloric acid

(G) Baking soda powder

24. Two straight lines cross. One of the four angles formed is 110 degrees. What are the other three angles? [Lines & Angles]

(A) 110, 90 and 70 degrees

(B) 110, 70 and 70 degrees

(C) 110, 110 and 110 degrees

(D) 70, 70 and 70 degrees

(E) 90, 90 and 90 degrees

(F) 110, 80 and 70 degrees

25. A thin nichrome wire glows red when current passes, but a thick copper wire joined the same way to the same cell barely warms up. Why does only the nichrome glow? [Electric Current & its Effects]

(A) Nichrome is thicker so it traps more current

(B) Copper has no electrons to carry current

(C) Nichrome contains a tiny hidden cell

(D) Copper is an insulator so nothing happens

(E) Nichrome has high resistance so it heats strongly and glows, while copper lets current pass easily without heating much

(F) Copper is magnetic and nichrome is not

(G) Nichrome stores electricity inside it

26. Write 18/-24 in standard form (lowest terms with a positive denominator). [Rational Numbers]

- **(A)** -3/4
- **(B)** 3/4
- **(C)** -9/12
- **(D)** -6/8
- **(E)** -2/3
- **(F)** 4/-3
- **(G)** -18/24

27. A colourless solution turns phenolphthalein PINK. What is its nature, and what will blue litmus do in it? [Acids, Bases & Salts]

- **(A)** It is a salt; blue litmus turns yellow
- **(B)** It is basic; blue litmus stays blue
- **(C)** It is neutral; blue litmus turns red
- **(D)** It is acidic; phenolphthalein is wrong
- **(E)** It is acidic; blue litmus turns red
- **(F)** It is basic; blue litmus turns red

28. When a transversal cuts two PARALLEL lines, each pair of corresponding angles is what? [Lines & Angles]

- **(A)** Adding to 180 degrees
- **(B)** Always 90 degrees
- **(C)** Adding to 90 degrees
- **(D)** Complementary
- **(E)** Supplementary
- **(F)** Equal
- **(G)** Always unequal

29. An electric heater is switched on for 3 hours each day and costs 9 rupees for every hour it runs. What is the total cost of running it through a 30-day month? [Electric Current & its Effects]

- **(A)** 90 rupees
- **(B)** 720 rupees
- **(C)** 900 rupees
- **(D)** 810 rupees
- **(E)** 270 rupees
- **(F)** 27 rupees
- **(G)** 108 rupees

30. Which of these is the GREATEST: -3/4, -2/3, -5/6, -1/2? [Rational Numbers]

- **(A)** they are all equal
- **(B)** -6/6
- **(C)** -5/6
- **(D)** -2/3
- **(E)** -4/4
- **(F)** -1/2
- **(G)** -3/4

31. A beaker holds 200 mL of acid. Adding a base neutralises 70% of the acid. How many millilitres of acid remain un-neutralised? [Acids, Bases & Salts]

- **(A)** 20 mL
- **(B)** 160 mL
- **(C)** 60 mL
- **(D)** 130 mL
- **(E)** 30 mL
- **(F)** 140 mL
- **(G)** 70 mL

32. A transversal cuts two parallel lines. One alternate interior angle measures 65 degrees. What is the co-interior (allied) angle on the same side as it? [Lines & Angles]

- **(A)** 90 degrees
- **(B)** 125 degrees
- **(C)** 65 degrees
- **(D)** 75 degrees
- **(E)** 105 degrees
- **(F)** 25 degrees
- **(G)** 115 degrees

33. A learner connects both ends of a single wire to the SAME terminal of a cell, with a bulb in the wire. Will the bulb light, and why? [Electric Current & its Effects]

- **(A)** No, because one wire is too short
- **(B)** Yes, but the bulb will burst
- **(C)** Yes, only if the wire is copper
- **(D)** Yes, one terminal is enough to push current
- **(E)** No, current needs a complete loop from one terminal of the cell to the other
- **(F)** No, because the bulb faces the wrong way
- **(G)** No, because the cell is too weak

34. Evaluate (-1/2) + (3/4) - (1/6) and give the answer in lowest terms. [Rational Numbers]

- **(A)** 7/12
- **(B)** 13/12
- **(C)** 11/12
- **(D)** 1/12
- **(E)** 5/12
- **(F)** 1/4
- **(G)** -1/12

35. Bacteria in the mouth make acids that attack tooth enamel. Most toothpastes are mildly basic. Why are they made basic? [Acids, Bases & Salts]

- **(A)** To turn litmus red
- **(B)** To feed the bacteria
- **(C)** To neutralise the mouth acids and protect the enamel
- **(D)** To make the teeth more acidic
- **(E)** To stain the teeth red
- **(F)** To make teeth taste sour

36. Three angles lie next to each other along a straight line at one point and measure x, 2x and 3x. What is x? [Lines & Angles]

- **(A)** 18 degrees
- **(B)** 45 degrees
- **(C)** 60 degrees
- **(D)** 90 degrees
- **(E)** 36 degrees
- **(F)** 30 degrees

37. A doorbell is pressed and held for 45 seconds. Its hammer strikes the gong 8 times every second while pressed. How many times does the hammer strike in all? [Electric Current & its Effects]

- **(A)** 53
- **(B)** 36
- **(C)** 320
- **(D)** 405
- **(E)** 8
- **(F)** 360
- **(G)** 400

38. The sum of a rational number and 5/8 is 1/4. What is that rational number? [Rational Numbers]

- **(A)** -3/8
- **(B)** 9/8
- **(C)** 3/8
- **(D)** 7/8
- **(E)** -1/8
- **(F)** -3/4
- **(G)** -7/8

39. A solution turns blue litmus RED, and red litmus shows NO change (stays red). What is the nature of this solution? [Acids, Bases & Salts]

- **(A)** Neutral
- **(B)** Acidic
- **(C)** Cannot be decided
- **(D)** Basic
- **(E)** Salt water only
- **(F)** Both acid and base

40. For two angles to be called ADJACENT angles, what must they share? [Lines & Angles]

- **(A)** A common vertex and a common arm
- **(B)** Both being 90 degrees
- **(C)** Only a common arm
- **(D)** Only a common vertex
- **(E)** Equal measures
- **(F)** Nothing at all
- **(G)** Only that they are equal in size

41. Two cells of 1.5 V are put in a holder, but one cell is placed backwards so its push opposes the other (negative facing negative). What is the effective voltage driving current in the circuit? [Electric Current & its Effects]

- **(A)** 3 V
- **(B)** 2.25 V
- **(C)** 0 V
- **(D)** 0.5 V
- **(E)** 0.75 V
- **(F)** 4.5 V
- **(G)** 1.5 V

42. Some rational number, when it is multiplied by -3/5, gives a product of -9/20. What is that number? [Rational Numbers]

- **(A)** -3/4
- **(B)** 3/5
- **(C)** 3/4
- **(D)** -27/100
- **(E)** 4/3
- **(F)** 9/20
- **(G)** 27/100

43. Which set correctly matches a sour food to the acid it contains? [Acids, Bases & Salts]

- **(A)** Curd - citric acid; vinegar - citric acid; lemon - lactic acid
- **(B)** Lemon - lactic acid; curd - acetic acid; vinegar - citric acid
- **(C)** Citrus fruits - citric acid; curd - lactic acid; vinegar - acetic acid
- **(D)** All three contain only hydrochloric acid
- **(E)** Curd - citric acid; lemon - acetic acid; vinegar - lactic acid
- **(F)** Citrus fruits - acetic acid; curd - citric acid; vinegar - lactic acid
- **(G)** Vinegar - lactic acid; curd - acetic acid; lemon - citric acid

44. The supplement of an angle is twice the angle itself. What is the angle? [Lines & Angles]

- **(A)** 30 degrees
- **(B)** 120 degrees
- **(C)** 60 degrees
- **(D)** 100 degrees
- **(E)** 90 degrees
- **(F)** 72 degrees

45. A faulty device that should pull 2 A suddenly starts pulling 4 A on a wire protected by a 3 A fuse. What does the fuse do, and why is this useful? [Electric Current & its Effects]

- **(A)** It stores the extra current for later
- **(B)** It melts and breaks the circuit, cutting off the dangerous extra current
- **(C)** It rings a bell but lets current pass
- **(D)** It carries the 4 A safely since it is only 1 A over
- **(E)** It cools the wire so the 4 A becomes safe
- **(F)** It increases the current to help the device

46. How many rational numbers lie strictly between 1/4 and 3/4? [Rational Numbers]

- **(A)** Exactly 4
- **(B)** Exactly 10
- **(C)** None
- **(D)** Exactly 2
- **(E)** Exactly 3
- **(F)** Exactly 1
- **(G)** Infinitely many

47. A dropper holds 60 drops of test solution, and each litmus test uses 4 drops. How many separate tests can be done with one full dropper? [Acids, Bases & Salts]

- **(A)** 240 tests
- **(B)** 12 tests
- **(C)** 15 tests
- **(D)** 64 tests
- **(E)** 56 tests
- **(F)** 20 tests
- **(G)** 30 tests

48. All the angles formed around a single point add up to how many degrees? [Lines & Angles]

- **(A)** 360 degrees
- **(B)** 60 degrees
- **(C)** 540 degrees
- **(D)** 720 degrees
- **(E)** 270 degrees
- **(F)** 180 degrees

49. A bulb normally runs on three cells of 1.5 V each in series. The shop only stocks 1.2 V cells. How many 1.2 V cells in series are needed to reach AT LEAST the original voltage? [Electric Current & its Effects]

- **(A)** 9 cells
- **(B)** 8 cells
- **(C)** 6 cells
- **(D)** 3 cells
- **(E)** 4 cells
- **(F)** 2 cells

50. A point on the number line starts at 2/3, moves LEFT by 7/6, then moves RIGHT by 1/2. Where does it end? [Rational Numbers]

- **(A)** 1/3
- **(B)** 0
- **(C)** 1/2
- **(D)** 2/3
- **(E)** -1
- **(F)** 1
- **(G)** -1/3

51. Which list of properties belongs to ACIDS? [Acids, Bases & Salts]

- **(A)** Sour taste and turn blue litmus red
- **(B)** Sweet taste and no effect on litmus
- **(C)** Sweet taste and turn turmeric green
- **(D)** Soapy feel and turn turmeric red
- **(E)** Soapy feel and turn blue litmus stay blue
- **(F)** Bitter taste and turn red litmus blue
- **(G)** Bitter taste and turn blue litmus red

52. Four angles meet around a point and measure 90 degrees, 100 degrees, x, and 80 degrees. What is x? [Lines & Angles]

- **(A)** 60 degrees
- **(B)** 70 degrees
- **(C)** 270 degrees
- **(D)** 90 degrees
- **(E)** 100 degrees
- **(F)** 80 degrees
- **(G)** 110 degrees

53. Which list contains ONLY devices that work mainly because of the heating effect of electric current? [Electric Current & its Effects]

- **(A)** Torch bulb, fan, fridge compressor
- **(B)** Electric bell, fan, radio
- **(C)** Electric iron, room heater, toaster
- **(D)** Computer, fan, doorbell
- **(E)** Loudspeaker, bell, motor
- **(F)** Television, bell, crane

54. What is the additive inverse of (-11/13)? [Rational Numbers]

- **(A)** 13/11
- **(B)** 11/26
- **(C)** -11/13
- **(D)** -13/11
- **(E)** 11/13
- **(F)** 0
- **(G)** 1

55. Soil that is too acidic for a crop is treated with quick lime, which is a base. A field needs 8 kg of lime per acre. How much lime is needed for a 5.5-acre field? [Acids, Bases & Salts]

- **(A)** 40 kg
- **(B)** 4.4 kg
- **(C)** 22 kg
- **(D)** 13.5 kg
- **(E)** 44 kg
- **(F)** 48 kg

56. Which of these pairs of angles is COMPLEMENTARY? [Lines & Angles]

(A) 25 degrees and 55 degrees
(B) 20 degrees and 80 degrees
(C) 40 degrees and 140 degrees
(D) 50 degrees and 50 degrees
(E) 60 degrees and 120 degrees
(F) 35 degrees and 55 degrees

57. An electric iron is run for 25 minutes to press shirts, and then twice as long to press bedsheets. What is the total time the iron runs? [Electric Current & its Effects]

(A) 1 hour 15 minutes
(B) 1 hour 25 minutes
(C) 2 hours 5 minutes
(D) 45 minutes
(E) 50 minutes
(F) 1 hour 5 minutes

58. Express -5/7 as an equal rational number whose denominator is 35. [Rational Numbers]

(A) -5/35
(B) -25/35
(C) -35/25
(D) 25/35
(E) -10/14
(F) -12/35

59. When an acid reacts completely with a base, what are the two products formed? [Acids, Bases & Salts]

(A) Two new acids
(B) Only water
(C) Only a salt
(D) A salt and water
(E) A stronger base
(F) An acid and a gas
(G) A salt and a gas

60. Two angles of a linear pair are in the ratio 5 : 7. What is the size of the BIGGER angle? [Lines & Angles]

(A) 84 degrees
(B) 70 degrees
(C) 60 degrees
(D) 105 degrees
(E) 75 degrees
(F) 90 degrees

61. An electromagnet made by winding wire on an iron nail holds several pins while current flows. The moment the current is switched off, the pins fall. Why do they fall? [Electric Current & its Effects]

- **(A)** The iron nail melts when current stops
- **(B)** The wire pulls the pins away
- **(C)** The pins become heavier when the current stops
- **(D)** The nail rusts instantly when switched off
- **(E)** The magnetism lasts only while current flows and vanishes when the current stops
- **(F)** The pins repel the nail after switch-off
- **(G)** The pins turn into permanent magnets themselves

62. Which is LARGER: the value of (2/3) x (-3/4), or the value of (-1/2) + (-1/12)? [Rational Numbers]

- **(A)** Both equal -1/2
- **(B)** They are equal
- **(C)** Both equal -7/12
- **(D)** Both are positive numbers
- **(E)** Negative numbers cannot be compared
- **(F)** (2/3) x (-3/4) is larger
- **(G)** (-1/2) + (-1/12) is larger

63. A pool is slightly acidic. Last week 3 scoops of a base were needed to bring it to neutral, where each scoop neutralises a fixed amount of acid. This week the acid is twice as much. How many scoops are needed now? [Acids, Bases & Salts]

- **(A)** 12 scoops
- **(B)** 6 scoops
- **(C)** 8 scoops
- **(D)** 3 scoops
- **(E)** 9 scoops
- **(F)** 1.5 scoops

64. When a transversal cuts two parallel lines, the co-interior (allied) angles are what? [Lines & Angles]

- **(A)** Always unequal and unrelated
- **(B)** Supplementary, adding to 180 degrees
- **(C)** Adding to 360 degrees
- **(D)** Equal to each other
- **(E)** Adding to 270 degrees
- **(F)** Complementary, adding to 90 degrees
- **(G)** Always exactly 90 degrees

65. A pupil uses a bulb-and-cell tester on 20 materials; the bulb lights up for 8 of them. What fraction of the materials are conductors, written in its lowest terms? [Electric Current & its Effects]

- **(A)** 1/5
- **(B)** 4/5
- **(C)** 2/4
- **(D)** 8/20
- **(E)** 2/10
- **(F)** 3/5
- **(G)** 2/5

66. Divide (-15/16) by (-5/8) and simplify. [Rational Numbers]

- **(A)** 120/80
- **(B)** 75/128
- **(C)** 3/2
- **(D)** 6/5
- **(E)** 2/3
- **(F)** 3/4
- **(G)** -3/2

67. China rose (gudhal) indicator is used to test two solutions. What colours does it give with an acid and with a base? [Acids, Bases & Salts]

- **(A)** Dark pink (magenta) with an acid and green with a base
- **(B)** Red with both acid and base
- **(C)** No change with either acid or base
- **(D)** Blue with an acid and red with a base
- **(E)** Pink with a base and green with an acid
- **(F)** Yellow with an acid and red with a base

68. An angle is 20 degrees less than its own supplement. What is the angle? [Lines & Angles]

- **(A)** 160 degrees
- **(B)** 110 degrees
- **(C)** 80 degrees
- **(D)** 70 degrees
- **(E)** 100 degrees
- **(F)** 90 degrees
- **(G)** 60 degrees

69. Which group lists materials that are ALL good conductors of electricity? [Electric Current & its Effects]

- **(A)** Copper wire, iron nail, aluminium foil
- **(B)** Glass rod, copper wire, chalk
- **(C)** Chalk, rubber, iron nail
- **(D)** Plastic, aluminium foil, dry paper
- **(E)** Iron nail, rubber, cotton thread
- **(F)** Wood, glass, copper wire
- **(G)** Rubber band, plastic ruler, dry wood

70. A diver is at -5/2 m (below sea level). She rises 3/4 m three separate times. What is her final position? [Rational Numbers]

- **(A)** -7/4 m
- **(B)** -19/4 m
- **(C)** 1/4 m
- **(D)** 7/4 m
- **(E)** -1/4 m
- **(F)** 1/2 m

71. Equal volumes of an acid (turns litmus red) and a base (turns litmus blue) are mixed, and the mixture now changes neither litmus paper. What happened, and what is mainly formed? [Acids, Bases & Salts]

- **(A)** Two gases were released
- **(B)** A new acid was made
- **(C)** The mixture turned strongly basic
- **(D)** They neutralised each other, forming salt and water
- **(E)** The base grew stronger
- **(F)** Nothing reacted at all
- **(G)** The acid grew stronger

72. A linear pair of angles is always what? [Lines & Angles]

- **(A)** Always equal to each other
- **(B)** Always complementary and equal
- **(C)** Supplementary and adjacent
- **(D)** Always two right angles
- **(E)** Adding to 360 degrees
- **(F)** Vertically opposite

73. Three bulbs in a series loop carry a current of 0.6 A. When one more identical bulb is added in the same series loop, the current falls to 0.45 A. By what fraction did the current fall? [Electric Current & its Effects]

- **(A)** 0.15
- **(B)** 1/3
- **(C)** 3/4
- **(D)** 1/4
- **(E)** 2/5
- **(F)** 1/2
- **(G)** 1/5

74. Find the product (-1/2) x (2/3) x (-3/5). [Rational Numbers]

- **(A)** 6/30
- **(B)** -1/5
- **(C)** 1/5
- **(D)** 3/10
- **(E)** 2/5
- **(F)** 1/30
- **(G)** -1/30

75. A flask is exactly neutral after acid and base react fully. Then 10 mL of extra acid is poured in. What is the nature of the mixture now? [Acids, Bases & Salts]

- **(A)** Cannot say
- **(B)** Basic
- **(C)** Water only
- **(D)** Acidic
- **(E)** Neutral
- **(F)** It stays exactly neutral forever
- **(G)** Salt only

76. A transversal cuts two parallel lines, making a corresponding pair where one angle is 3x and the other is x + 50. What is x? [Lines & Angles]

- **(A)** 20
- **(B)** 75
- **(C)** 30
- **(D)** 100
- **(E)** 12.5
- **(F)** 50
- **(G)** 25

77. In an electric bell, which part repeatedly MAKES and BREAKS the circuit so that the hammer keeps striking the gong? [Electric Current & its Effects]

- **(A)** The sound waves travelling in the air
- **(B)** The cell inside the bell
- **(C)** The gong itself
- **(D)** The push button only
- **(E)** The coil of insulated wire
- **(F)** The contact screw touching the springy strip (armature)

78. Between which two consecutive integers does -17/5 lie? [Rational Numbers]

- **(A)** -5 and -4
- **(B)** 0 and -1
- **(C)** -2 and -1
- **(D)** -3 and -2
- **(E)** 3 and 4
- **(F)** -17 and -5
- **(G)** -4 and -3

79. Milk of magnesia is a mild base taken as a medicine for acidity. How does it relieve an acidic stomach? [Acids, Bases & Salts]

- **(A)** It coats the stomach with oil
- **(B)** It turns the acid into a gas only
- **(C)** It makes the stomach more sour
- **(D)** It washes the acid out with plain water
- **(E)** It neutralises the excess acid in the stomach
- **(F)** It cools the stomach with ice
- **(G)** It adds more acid to the stomach

80. At a road crossing, two straight roads meet so that one angle A is 125 degrees. Let B be the angle vertically opposite A, and C the angle adjacent to A on a straight line. What is B + C? [Lines & Angles]

- **(A)** 250 degrees
- **(B)** 180 degrees
- **(C)** 235 degrees
- **(D)** 305 degrees
- **(E)** 110 degrees
- **(F)** 90 degrees

81. A toy needs a 6 V battery of four 1.5 V cells in series. Two of the cells leak and now give only 1 V each, while the other two stay at 1.5 V. What is the total voltage of the battery now? [Electric Current & its Effects]

- **(A)** 3 V
- **(B)** 5 V
- **(C)** 6 V
- **(D)** 5.5 V
- **(E)** 4 V
- **(F)** 2 V

82. If (x/4) + (1/3) = -1/6, what is the value of x? [Rational Numbers]

- **(A)** -2
- **(B)** -8
- **(C)** -1/8
- **(D)** -1/2
- **(E)** 2
- **(F)** 8

83. A bottle of vinegar is 5% acetic acid by volume. In 250 mL of this vinegar, how many millilitres are pure acetic acid? [Acids, Bases & Salts]

- **(A)** 20 mL
- **(B)** 12.5 mL
- **(C)** 50 mL
- **(D)** 1.25 mL
- **(E)** 237.5 mL
- **(F)** 5 mL
- **(G)** 25 mL

84. In a figure, angle 1 and angle 2 are complementary, and angle 2 and angle 3 are supplementary. If angle 1 is 28 degrees, what is angle 3? [Lines & Angles]

- **(A)** 90 degrees
- **(B)** 122 degrees
- **(C)** 98 degrees
- **(D)** 28 degrees
- **(E)** 62 degrees
- **(F)** 118 degrees
- **(G)** 152 degrees

85. Which statement about an MCB (miniature circuit breaker) is correct? [Electric Current & its Effects]

- **(A)** It turns torch cells into bulbs
- **(B)** It stores electricity to use later
- **(C)** It switches off the circuit on overload and can be switched back on after the fault is fixed
- **(D)** It must be replaced with fresh wire every time it trips
- **(E)** It makes bulbs glow brighter
- **(F)** It boosts the current to run heavy appliances

86. Simplify (3/4) - (5/6) + (1/12). [Rational Numbers]

- **(A)** -1/6
- **(B)** 1/4
- **(C)** 0
- **(D)** 1/2
- **(E)** 1/12
- **(F)** -1/12
- **(G)** 1/6

87. From which living things is the common laboratory indicator litmus obtained? [Acids, Bases & Salts]

- **(A)** Turmeric roots
- **(B)** Lichens
- **(C)** Sea salt crystals
- **(D)** Coal lumps
- **(E)** Rose flowers
- **(F)** Bones of animals
- **(G)** Leaves of any plant

88. Two angles are both supplementary to each other AND equal in size. What is each angle? [Lines & Angles]

- **(A)** 180 degrees
- **(B)** 90 degrees
- **(C)** 60 degrees
- **(D)** 120 degrees
- **(E)** 100 degrees
- **(F)** 45 degrees
- **(G)** 30 degrees

89. A heater warms a room by 2 C every 5 minutes at a steady rate. The room is at 16 C and you want it at 24 C. For how long must the heater run? [Electric Current & its Effects]

- **(A)** 20 minutes
- **(B)** 10 minutes
- **(C)** 12 minutes
- **(D)** 16 minutes
- **(E)** 24 minutes
- **(F)** 25 minutes

90. Divide (-7/2) by (7/4). Is the result a whole-number integer, and what is it? [Rational Numbers]

- **(A)** No, it is -1/2
- **(B)** No, it is 1/2
- **(C)** Yes, it is -4
- **(D)** Yes, it is -2
- **(E)** No, it is 4
- **(F)** No, it is -49/8

91. To test 9 different solutions, each tested twice, a student uses one fresh strip of blue litmus per test. How many strips of blue litmus are needed in all? [Acids, Bases & Salts]

- **(A)** 9 strips
- **(B)** 11 strips
- **(C)** 18 strips
- **(D)** 27 strips
- **(E)** 36 strips
- **(F)** 20 strips

92. The angles of a linear pair are (2x + 10) degrees and (3x + 20) degrees. Find x and the LARGER angle. [Lines & Angles]

- **(A)** x = 25, larger angle = 95 degrees
- **(B)** x = 30, larger angle = 80 degrees
- **(C)** x = 30, larger angle = 110 degrees
- **(D)** x = 20, larger angle = 100 degrees
- **(E)** x = 30, larger angle = 70 degrees
- **(F)** x = 40, larger angle = 90 degrees

93. A wire carrying current is held over a compass; the needle swings aside. When the current is switched off, what happens to the compass needle? [Electric Current & its Effects]

- **(A)** It points straight up
- **(B)** It spins round and round forever
- **(C)** It melts from the heat
- **(D)** It keeps pointing in the swung-aside direction
- **(E)** It glows red hot and stays put
- **(F)** It swings back to point along the north-south direction
- **(G)** It snaps in two

94. For any non-zero rational number, what is the product of the number and its reciprocal? [Rational Numbers]

- **(A)** 0
- **(B)** -1
- **(C)** Its square
- **(D)** 1
- **(E)** Undefined
- **(F)** Always 100
- **(G)** The number itself

95. Why must acidic waste from a factory be neutralised before it is let into a river? [Acids, Bases & Salts]

- **(A)** Acidic water harms fish, plants and people who use the river
- **(B)** Acidic water cleans the river automatically
- **(C)** Acidic water has no effect on living things
- **(D)** Acidic water makes fish grow bigger
- **(E)** Acidic water turns into drinking water
- **(F)** Acidic water makes the river flow faster
- **(G)** Acidic water makes the river smell sweet

96. Which of these statements is FALSE? [Lines & Angles]

(A) Two acute angles can form a linear pair
(B) Angles around a point add up to 360 degrees
(C) Adjacent angles share a common arm
(D) Complementary angles add up to 90 degrees
(E) Two right angles can form a linear pair
(F) Vertically opposite angles are equal

97. An immersion rod heats 3 L of water in 9 minutes. To fill a bucket you need to heat 8 L the same way. If the time needed is proportional to the amount of water, how long will it take? [Electric Current & its Effects]

(A) 14 minutes
(B) 11 minutes
(C) 24 minutes
(D) 72 minutes
(E) 27 minutes
(F) 21 minutes

98. A recipe needs 3/4 cup of sugar. You decide to make 2 and 1/2 times the recipe. How much sugar do you need? [Rational Numbers]

(A) 3 and 1/8 cups
(B) 2 and 1/4 cups
(C) 2 and 7/8 cups
(D) 1 and 1/4 cups
(E) 6/8 cup
(F) 1 and 1/2 cups
(G) 1 and 7/8 cups

99. Three solutions are tested. P turns blue litmus red. Q turns turmeric paper red. R changes neither litmus paper. Which solution is basic? [Acids, Bases & Salts]

(A) P and Q
(B) P and R
(C) None of them
(D) P
(E) Q
(F) R
(G) All three

100. A transversal cuts two parallel lines. On one side, the interior angle is 3 times the co-interior angle next to it. What is the SMALLER of these two angles? [Lines & Angles]

(A) 135 degrees
(B) 90 degrees
(C) 45 degrees
(D) 30 degrees
(E) 75 degrees
(F) 36 degrees